PROJECT LOGBOOK

Gap-Aware Retrieval (GAR): Adaptive Retrieval Using Information Gaps and Existing Evidence

Mini Project - Bachelor of Technology in Computer Engineering

Period covered: 15 July 2025 to 31 August 2025

Kunal Dahal (22CSEC17)

Aniket Manger (22CSEC23)

Suman Chettri (22CSEC44)

Under the supervision of

Mr. Anmol Pandey, Assistant Professor

Department of Computer Engineering

Sikkim Institute of Science and Technology

(A constituent college of Sikkim University)

PART A - DAILY LOG (15 JULY 2025 - 31 AUGUST 2025)

Day 1 - 15 July 2025 - Project setup and area scoping

Work done: Fixed the broad problem area for the mini project after a group discussion and a meeting with the supervisor. The area selected is Retrieval-Augmented Generation (RAG), and within it, the retrieval stage of multi-step question answering. No paper was studied in depth today.

Setup completed: A shared folder for papers, a reading sheet with fixed columns (paper, authors, venue, year, findings, research gap, relevance to our work, decision), an Overleaf project for the report, and this logbook.

Reading rule fixed for the whole study phase: Every paper read from this point is recorded in the same four parts - what it does, what it does not do (gap), how it relates to our work, and whether it is carried forward or dropped. Papers that turn out to be of low relevance are still recorded in one or two lines and are still listed in the reference section.

Outcome: Study plan agreed. Literature study starts from Day 2, base framework selection targeted within the first week.

Day 2 - 16 July 2025 - Background reading on Retrieval-Augmented Generation

Papers studied: "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks" by P. Lewis et al., NeurIPS 2020 [1]; "REALM: Retrieval-Augmented Language Model Pre-Training" by K. Guu et al., ICML 2020 [2]; "Leveraging Passage Retrieval with Generative Models for Open Domain Question Answering" by G. Izacard and E. Grave, EACL 2021 [3]; and the survey "Retrieval-Augmented Generation for Large Language Models: A Survey" by Y. Gao et al., 2023 [4].

Findings:

- RAG connects a retriever to a generator so that the language model can answer using external documents instead of only its stored parameters [1].
- REALM shows that retrieval can be learned together with the language model during pre-training, which establishes retrieval as a trainable part of the system rather than a fixed pre-processing step [2].
- Fusion-in-Decoder shows that answer quality keeps improving as more retrieved passages are provided to the reader, which means the reader depends heavily on what the retriever supplies [3].
- The survey groups RAG work into three stages - pre-retrieval, retrieval, and post-retrieval - and lists iterative and adaptive retrieval as an active research direction [4].

Research gap noted:

- All three systems retrieve using the original query only. None of them changes the retrieval objective after some evidence has already been collected.

Relevance to our work:

- These papers give the standard pipeline that our work sits inside, and they confirm that the retrieval stage sets an upper limit on the final answer quality. This is why we chose to work on retrieval rather than on generation.

Outcome: The general RAG pipeline is understood. Next step is to select one iterative framework as the base for the project.

Day 3 - 17 July 2025 - Base paper: MIGRES (first complete read)

Paper studied: "LLMs Know What They Need: Leveraging a Missing Information Guided Framework to Empower Retrieval-Augmented Generation" by K. Wang, F. Duan, P. Li, S. Wang, and X. Cai, COLING 2025, pp. 2379-2400 [5].

Why this paper: It is the only framework found so far that makes the missing information an explicit, readable state of the system instead of hiding it inside the model's reasoning. This makes it possible for another component to use that state.

Findings:

- The paper shows experimentally that an LLM can both extract useful information from retrieved documents and state what information is still missing.
- MIGRES stands for Missing Information Guided Retrieve-Extraction-Solving. It is an iterative loop: identify what is missing, generate a targeted query from the missing information, retrieve, extract useful knowledge, update the state, and repeat until the question can be answered.
- Sentence-level reranking and filtering are applied to the retrieved documents before extraction, to remove irrelevant sentences.
- The framework reports improvements over single-step and other iterative RAG baselines on multi-hop question answering.

Research gap:

- Performance still depends on the quality of the generated query and on what the retriever returns for that query. The missing information is used only to write a new query string; it is not used inside the ranking function.
- Hard-negative passages that score high on relevance but carry no useful information can still enter the context.

Relevance to our work:

- MIGRES becomes the base framework of the project. It gives us, at every iteration, exactly the three things a retriever would need: the original question, the evidence already collected, and the information still missing.

Outcome: MIGRES selected as the base paper. Next day allotted to its implementation.

Day 4 - 18 July 2025 - MIGRES: repository and architecture study

Work done: Went through the official code released with the paper [5] and mapped the modules of the pipeline: question processing, missing-information generation, query generation, retrieval call, knowledge filtering, information extraction, answer solving, and the

stopping condition.

Points recorded:

- The retrieval call is a separate module. The reasoning modules communicate with it only through a query string and a returned document list, so the retriever can be replaced without touching the rest of the framework.
- At the point where retrieval is called, the framework already holds the question, the accumulated evidence, and the current missing-information list in memory. These are available to any replacement retriever.
- The retrieval baseline used in the framework is a sparse lexical retriever, so any new retriever must be compared against it first.

Open item recorded: The backbone LLM used by MIGRES for gap identification, extraction and solving must be confirmed from the experimental section of the paper and from the configuration file in the repository, and then written explicitly into our methodology. Until this is confirmed, our methodology cannot claim that the surrounding framework is held fixed. This item is carried to Part B and is still open.

Outcome: Confirmed that the retrieval component is the replaceable part of MIGRES.

Day 5 - 19 July 2025 - MIGRES: gap analysis and project direction

Work done: Wrote down, in one page, what MIGRES does well, what it leaves unused, and where our contribution can sit.

Observations:

- MIGRES computes the missing information and then throws most of it away: only its text form is used to build a query. The ranking function of the retriever never sees it.
- The evidence already collected is used by the reasoning modules but is never used to penalise documents that repeat what is already known.
- Therefore the retrieval stage of MIGRES is still a single-query relevance ranker running inside an iterative loop.

Direction fixed: Keep the whole MIGRES framework unchanged and replace only the retrieval algorithm with one that reads the current information state. Working name for the proposed algorithm: Gap-Aware Retrieval (GAR).

Outcome: Project direction fixed. The next block of days is used to study retrieval algorithms one by one, since our contribution is a retrieval algorithm and it must be compared against retrieval algorithms.

Day 6 - 20 July 2025 - Information retrieval basics and BM25

Paper studied: "The Probabilistic Relevance Framework: BM25 and Beyond" by S. Robertson and H. Zaragoza, Foundations and Trends in Information Retrieval, 2009 [6].

Findings:

- BM25 scores a document using term frequency, inverse document frequency, and document-length normalisation, with parameters k1 and b.

- It is a sparse lexical method: a document is scored only on the query terms it literally contains.
- It remains a strong baseline and is cheap to run, which is why it is widely used as the first-stage retriever.

Research gap:

- No semantic matching. A document that answers the question in different words is not retrieved.
- The score depends only on the query. There is no notion of what the system already knows.

Relevance to our work:

- BM25 is the retrieval baseline inside MIGRES, so MIGRES+BM25 is our main comparison system and the reference point for the primary hypothesis.

Outcome: BM25 confirmed as main baseline.

Day 7 - 21 July 2025 - Dense Passage Retrieval (DPR)

Paper studied: "Dense Passage Retrieval for Open-Domain Question Answering" by V. Karpukhin et al., EMNLP 2020 [7].

Findings:

- DPR uses two separate BERT encoders, one for questions and one for passages, and ranks passages by the dot product of their dense vectors.
- Training uses in-batch negatives and hard negatives obtained from BM25.
- Dense retrieval outperforms BM25 on open-domain QA when supervised training data is available.

Research gap:

- DPR needs labelled question-passage pairs, so its quality depends on the training domain.
- The question encoder takes a single query string. There is no place in the model where an evidence set or a missing-information set could be supplied.

Relevance to our work:

- DPR is included as the supervised dense baseline. It also confirms the design decision that our method should not require training data, because we cannot produce labelled gap-document pairs.

Outcome: DPR added to the baseline list.

Day 8 - 22 July 2025 - Contriever

Paper studied: "Unsupervised Dense Information Retrieval with Contrastive Learning" by G. Izacard et al., 2022 [8].

Findings:

- Contriever trains a dense retriever with contrastive learning on unlabelled text, using independently cropped spans of the same document as positive pairs.
- It performs competitively without any retrieval supervision and transfers reasonably across domains.

Research gap:

- Like DPR, it produces one vector per query and one per document, so the ranking decision is a single similarity value and cannot express "useful given what is already known".

Relevance to our work:

- Contriever is included as the unsupervised dense baseline, which makes the comparison fair with respect to training data: GAR uses no retrieval supervision either.

Note: The exact publication venue of this paper must be checked before the final reference list is frozen; it has appeared both as a preprint and in a journal track.

Outcome: Contriever added to the baseline list.

Day 9 - 23 July 2025 - ColBERT

Paper studied: "ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT" by O. Khattab and M. Zaharia, SIGIR 2020 [9].

Findings:

- Instead of compressing a document into one vector, ColBERT keeps one vector per token and scores a pair by summing, over query tokens, the maximum similarity against document tokens.
- This late-interaction design keeps most of the accuracy of a cross-encoder while allowing documents to be encoded offline.

Research gap:

- Token-level storage is expensive for large corpora.
- The scoring function is still defined between one query and one document.

Relevance to our work:

- Read mainly as background for ColBERTv2, and as evidence that a scoring function can be built from several similarity terms instead of a single dot product.

Outcome: Carried forward as background.

Day 10 - 24 July 2025 - ColBERTv2

Paper studied: "ColBERTv2: Effective and Efficient Retrieval via Lightweight Late Interaction" by K. Santhanam, O. Khattab, J. Saad-Falcon, C. Potts, and M. Zaharia, NAACL 2022, pp. 3715-3734 [10].

Findings:

- ColBERTv2 adds residual compression to reduce the storage cost of token-level vectors.
- Denoised supervision is used during training to reduce the effect of false negatives.
- It reaches strong retrieval quality at a small fraction of the original storage.

Research gap:

- Storage is still larger than single-vector dense retrieval.
- The architecture is designed for query-document matching and does not consider an information requirement that changes between retrieval steps.

Relevance to our work:

- Included as the multi-vector, late-interaction baseline. It is the strongest non-iterative retriever in our comparison set, so beating it is the harder part of the second hypothesis.

Outcome: ColBERTv2 added to the baseline list.

Day 11 - 25 July 2025 - SPLADE

Paper studied: "SPLADE: Sparse Lexical and Expansion Model for First Stage Ranking" by T. Formal, B. Piwowarski, and S. Clinchant, SIGIR 2021 [11].

Findings:

- SPLADE learns a sparse representation over the vocabulary, so each document is expanded with related terms and the result can still be served by an inverted index.
- A regularisation term controls how sparse the representation stays, which controls the efficiency-quality trade-off.

Research gap:

- The expansion is learned from the corpus and the query only; it does not depend on the state of an ongoing retrieval session.

Relevance to our work:

- Included as the neural sparse baseline, so that the comparison covers lexical, dense, late-interaction and neural-sparse families.

Outcome: SPLADE added to the baseline list. Baseline family coverage is now complete.

Day 12 - 26 July 2025 - Cross-encoder reranking (background, low relevance)

Papers skimmed: "Passage Re-ranking with BERT" by R. Nogueira and K. Cho, 2019 [12]; "Document Ranking with a Pretrained Sequence-to-Sequence Model" by R. Nogueira et al., Findings of EMNLP 2020 [13].

Short note: Both papers rerank a candidate list by feeding the query and the document together into a transformer, which is accurate but requires one model call per candidate document.

Decision: Not used. GAR must score a large candidate pool cheaply and without one model call per document, so cross-encoder reranking is outside our design. Recorded here and listed in the references only.

Day 13 - 27 July 2025 - Sentence embeddings and choice of embedding model

Papers studied: "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks" by N. Reimers and I. Gurevych, EMNLP 2019 [14]; the model card for multi-qa-mpnet-base-cos-v1 [15].

Findings:

- Sentence-BERT produces sentence-level embeddings that can be compared with cosine similarity, making large-scale semantic comparison practical.
- The multi-qa-mpnet-base-cos-v1 model produces 768-dimensional embeddings and is trained for semantic search, which matches our use case.

Relevance to our work:

- This fixes the representation function used by GAR. The same model will be used for the question, each gap requirement, each document and each evidence item, so that all three GAR components are computed in one common space.
- Because the model is used as-is and is not fine-tuned, GAR has no training stage. This is a deliberate design decision and must be stated in the methodology.

Outcome: Embedding model fixed for the primary configuration.

Day 14 - 28 July 2025 - Approximate nearest neighbour search

Papers studied: "Billion-Scale Similarity Search with GPUs" by J. Johnson, M. Douze, and H. Jegou, 2017 [16]; "Efficient and Robust Approximate Nearest Neighbor Search Using Hierarchical Navigable Small World Graphs" by Y. A. Malkov and D. A. Yashunin, IEEE TPAMI, 2020 [17].

Findings:

- ANN search returns approximate nearest vectors in sub-linear time, trading a small amount of recall for a large speed gain.
- HNSW builds a multi-layer proximity graph and is a common index choice for high-dimensional embeddings.

Relevance to our work:

- GAR cannot afford to score every document in the corpus, so ANN is used to build a candidate pool of size K0, and the full GAR score is computed only on that pool.
- Important distinction recorded: ANN here means approximate nearest neighbour, not artificial neural network. This confusion appeared in an early draft and was corrected.
- The candidate pool must be built from the question and from each gap requirement, otherwise a document that answers only the gap would never reach the scoring stage.

Outcome: Two-stage design fixed - ANN candidate generation followed by GAR scoring.

Day 15 - 29 July 2025 - Diversity in ranking: Maximal Marginal Relevance

Paper studied: "The Use of MMR, Diversity-Based Reranking for Reordering Documents and Producing Summaries" by J. Carbonell and J. Goldstein, SIGIR 1998 [18].

Findings:

- MMR selects the next document by combining its relevance to the query with its dissimilarity to the documents already selected, using a single trade-off parameter.

Research gap:

- The redundancy term is computed against the documents selected inside the same retrieval step. It does not consider evidence collected in earlier iterations of an outer loop, and it has no notion of an explicit information requirement.

Relevance to our work:

- This is the closest classical idea to our novelty component and it must be acknowledged as prior art. The difference to state clearly in the report is that MMR measures redundancy against the current result set, while GAR measures novelty against the accumulated evidence of the framework.

Outcome: Recorded as the reference point for the novelty term.

Day 16 - 30 July 2025 - Retrieval benchmarking (low relevance)

Paper skimmed: "BEIR: A Heterogeneous Benchmark for Zero-Shot Evaluation of Information Retrieval Models" by N. Thakur, N. Reimers, A. Ruckle, A. Srivastava, and I. Gurevych, NeurIPS Datasets and Benchmarks, 2021 [19].

Short note: BEIR is a zero-shot benchmark across many retrieval tasks and is useful for comparing general-purpose retrievers.

Decision: Not used for evaluation, because our evaluation must run inside MIGRES on multi-hop QA datasets where an information gap exists. Listed in the references only.

Day 17 - 31 July 2025 - Consolidation of the retrieval study

Work done: Wrote a comparison table of everything studied in Days 6 to 16, with columns: method, representation type, what the score depends on, training requirement, and whether the current information state can enter the score.

Conclusion recorded: In every method studied, the score is a function of the query and the document only. None of them takes the collected evidence or the missing information as an input. This is the concrete gap our work addresses, and it is a gap in the retrieval algorithms themselves, not only in MIGRES.

Baseline list frozen: BM25 [6], DPR [7], Contriever [8], ColBERTv2 [10], SPLADE [11]. ColBERT [9], the cross-encoder rerankers [12][13], and BEIR [19] are background only.

Outcome: Retrieval study block closed. Next block: frameworks comparable to MIGRES.

Day 18 - 1 August 2025 - Self-RAG

Paper studied: "Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection" by A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, ICLR 2024 [20].

Findings:

- The model is trained to emit reflection tokens that decide when to retrieve and that criticise whether a retrieved passage is relevant and whether the generated text is supported by it.
- Retrieval becomes on-demand rather than fixed, and passages are judged after retrieval.

Research gap:

- The critique happens after retrieval. The retriever itself is unchanged and still ranks by query relevance, so a useless passage must first be retrieved before it can be rejected.

- The method requires training the generator with the special tokens, which is expensive.

Relevance to our work:

- Confirms that the field repairs retrieval mistakes downstream instead of preventing them at the ranking stage. GAR acts one stage earlier, which is the argument for our contribution.

Outcome: Carried forward as a related framework, not as a baseline.

Day 19 - 2 August 2025 - Adaptive-RAG

Paper studied: "Adaptive-RAG: Learning to Adapt Retrieval-Augmented Large Language Models through Question Complexity" by S. Jeong, J. Baek, S. Cho, S. J. Hwang, and J. C. Park, NAACL 2024 [21].

Findings:

- A classifier predicts the complexity of a question and routes it to no retrieval, single-step retrieval, or multi-step retrieval.
- This reduces cost on easy questions while keeping accuracy on hard ones.

Research gap:

- The adaptation is at the level of the strategy, not the ranking function. Once the multi-step path is chosen, each retrieval step still ranks documents by query relevance.

Relevance to our work:

- Supports the claim that adaptivity in iterative RAG has been studied at the control level but not inside the retrieval score. It is also a candidate framework for a future generalisation experiment, to show that GAR is not tied to MIGRES.

Outcome: Recorded as a possible second host framework for a later generalisation study.

Day 20 - 3 August 2025 - IRCoT

Paper studied: "Interleaving Retrieval with Chain-of-Thought Reasoning for Knowledge-Intensive Multi-Step Questions" by H. Trivedi, N. Balasubramanian, T. Khot, and A. Sabharwal, ACL 2023 [22].

Findings:

- Retrieval and chain-of-thought reasoning are interleaved: each reasoning sentence is used as the query for the next retrieval step, and each retrieval informs the next reasoning sentence.
- It improves multi-hop QA over one-shot retrieval on several datasets.

Research gap:

- The reasoning step is used only to produce a query string. The retriever receives a string and nothing else, so the accumulated context is again invisible to the ranking function.

Relevance to our work:

- IRCoT and MIGRES share the same limitation, which strengthens the motivation. It also shows that the "state to string" bottleneck is the common pattern in iterative RAG.

Outcome: Recorded as a related framework.

Day 21 - 4 August 2025 - Self-Ask and ReAct

Papers studied: "Measuring and Narrowing the Compositionality Gap in Language Models" by O. Press et al., Findings of EMNLP 2023 [23]; "ReAct: Synergizing Reasoning and Acting in Language Models" by S. Yao et al., ICLR 2023 [24].

Findings:

- Self-Ask makes the model decompose a question into explicit follow-up questions and answer them one at a time, optionally with a search engine.
- ReAct interleaves reasoning traces with actions such as search, so the model decides what to look up next.

Research gap:

- Both treat the retriever as an external black box called with a string. Neither controls how the returned documents are ranked.

Relevance to our work:

- These two papers are the clearest statement of the decomposition idea that MIGRES formalises as a missing-information state. They are useful in the introduction to explain why a gap state exists at all.

Outcome: Recorded as background for the introduction.

Day 22 - 5 August 2025 - FLARE (active retrieval)

Paper studied: "Active Retrieval Augmented Generation" by Z. Jiang et al., EMNLP 2023 [25].

Findings:

- FLARE predicts the upcoming sentence, and if the model is not confident about it, that sentence is used as a query to retrieve before generating it.
- Retrieval is therefore triggered by the model's own uncertainty.

Research gap:

- Uncertainty decides when to retrieve, not which document is most useful. The ranking function remains a plain relevance ranker.

Relevance to our work:

- Useful contrast for the report: FLARE is adaptive in timing, Adaptive-RAG is adaptive in strategy, and GAR is adaptive in scoring. These three axes together make the positioning of our contribution clear.

Outcome: Recorded as a related framework.

Day 23 - 6 August 2025 - Iter-RetGen and DSP (brief)

Papers studied: "Enhancing Retrieval-Augmented Large Language Models with Iterative Retrieval-Generation Synergy" by Z. Shao et al., Findings of EMNLP 2023 [26]; "Demonstrate-Search-Predict" by O. Khattab et al., 2022 [27].

Short notes:

- Iter-RetGen uses the model's previous output as the next query, so generation and retrieval improve each other over iterations [26].

- DSP composes retrieval and language-model calls into programmable pipelines [27].

Decision: Both are recorded as related work in the introduction only. Neither changes the retrieval score, and neither is used as a baseline. Listed in the references.

Day 24 - 7 August 2025 - Framework comparison and confirmation of the base framework

Work done: Compared MIGRES [5] with Self-RAG [20], Adaptive-RAG [21], IRCoT [22], FLARE [25] and Iter-RetGen [26] on four points: does the framework expose an explicit information-gap state, does it expose accumulated evidence, can the retriever be replaced without retraining anything, and is code available.

Conclusion recorded: MIGRES is the only framework among these that exposes both an explicit missing-information state and accumulated evidence to the point in the pipeline where retrieval is called, and it allows the retriever to be swapped without retraining the generator. It is therefore confirmed as the experimental host framework.

Statement fixed for the report: MIGRES is the experimental framework and is not our contribution. GAR is the retrieval component and is our contribution. The main experimental variable is the retrieval algorithm; everything else is held fixed.

Outcome: Base framework confirmed. Next block: papers that specifically deal with usefulness, gain, coverage and redundancy of retrieved information, which are the ideas our score is built from.

Day 25 - 8 August 2025 - Relevant Information Gain and the Dartboard method

Paper studied: "Better RAG using Relevant Information Gain" by M. Pickett, J. Hartman, A. K. Bhowmick, R. ul Alam, and A. Vempaty, arXiv preprint arXiv:2407.12101, 2024 [28].

Findings:

- The limited context window means that redundant passages waste space that could carry new information.
- A Relevant Information Gain metric measures the total query-relevant information contained in a set of passages, rather than scoring each passage alone.
- The Dartboard retrieval method optimises this metric directly, and diversity appears as a consequence instead of being added as a separate penalty term.
- Reported to outperform relevance-based and diversity-based retrieval on the Retrieval Augmented Generation Benchmark.

Research gap:

- The objective is defined for selecting a set of passages for one query in one step. The paper does not define how the objective should change once evidence has been collected across several retrieval steps.
- Evaluation is mainly on one benchmark, so behaviour in iterative settings is left open.

Relevance to our work:

- This is the direct motivation for the novelty component of GAR. It establishes that redundancy is a real cost, and it also shows the limitation we extend: redundancy measured within one step, versus redundancy measured against the accumulated evidence of an iterative framework.

Outcome: Novelty component justified from the literature.

Day 26 - 9 August 2025 - InfoGain-RAG

Paper studied: "InfoGain-RAG: Boosting Retrieval-Augmented Generation through Document Information Gain-based Reranking and Filtering" by Z. Wang, Z. Liang, Z. Shao, Y. Ma, H. Dai, B. Chen, L. Mao, C. Lei, Y. Ding, and H. Li, EMNLP 2025, pp. 7190-7204 [29].

Findings:

- Document Information Gain measures how much one retrieved document contributes to producing the correct answer.
- It is computed from the change in the generation confidence of the LLM when the document is added to the context.
- These scores are used to train a multi-task reranker that both filters and ranks documents.
- Reported improvements across several RAG models and benchmarks.

Research gap:

- Evaluated only on text retrieval; other modalities are left for future work.
- The metric measures contribution to the answer but cannot tell whether the content of the document is factually correct.
- Reranker size is limited by compute, which may cap the achievable quality.

Relevance to our work:

- Confirms that "contribution" is a better ranking target than "relevance", which is the core assumption behind GAR.
- Important difference to state in the report: their gain is measured with respect to the answer and requires an LLM forward pass per document; our coverage and novelty terms are measured with respect to the gap and the evidence and are computed locally, without any LLM call per candidate.

Outcome: Design decision recorded - GAR scoring must not require one LLM call per candidate document.

Day 27 - 10 August 2025 - Iterative alignment-based evidence retrieval

Paper studied: "Unsupervised Alignment-Based Iterative Evidence Retrieval for Multi-Hop Question Answering" by V. Yadav, S. Bethard, and M. Surdeanu, ACL 2020, pp. 4514-4525 [30].

Findings:

- Evidence sentences are retrieved iteratively without supervision, using alignment between question terms and sentence terms.

- At each step the method explicitly tracks which parts of the question are already covered by the justification collected so far, and directs the next retrieval towards the parts that are not yet covered.
- A simple, parameter-light method is shown to be competitive with trained retrievers on multi-hop QA.

Research gap:

- The coverage is computed over question terms at sentence level and within one dataset setting. It is not formulated as a general document-ranking score, and it is not connected to an LLM-generated missing-information state.

Relevance to our work:

- This is the direct motivation for the coverage component of GAR, and it is important evidence that a term-level coverage measure works in practice without training.
- It also supports our choice of normalised token recall for coverage rather than a second embedding similarity, because coverage should measure how much of the requirement is actually present in the document.

Outcome: Coverage component justified from the literature. All three GAR components now have a source: gap relevance from MIGRES [5], coverage from [30], novelty from [28].

Day 28 - 11 August 2025 - Retrieval-augmented generation with source reliability

Paper studied: "Retrieval-Augmented Generation with Estimation of Source Reliability" by J. Hwang, J. Park, H. Park, D. Kim, S. Park, and J. Ok, EMNLP 2025, pp. 34279-34303 [31].

Findings:

- In multi-source RAG, ranking by query-document relevance alone allows unreliable sources to influence the answer.
- RA-RAG estimates the reliability of each source by cross-checking claims across sources, then retrieves selectively from reliable and relevant sources and aggregates with weighted majority voting.
- Performance improves over the baselines and holds as the number of sources grows.

Research gap:

- Reliability estimation is harder when few sources are available for cross-checking.
- Real-world evaluation focuses on news-type claims; specialised topics remain difficult.
- The estimation is offline and must be refreshed when source reliability changes.

Relevance to our work:

- This paper is the precedent for adding a non-relevance factor into document selection. It shows that a retrieval score can legitimately be a weighted combination of several signals, which is exactly the form of the GAR score.

Outcome: The weighted-combination form of the GAR score is supported by prior work.

Day 29 - 12 August 2025 - Context compression and filtering (brief)

Papers skimmed: "RECOMP: Improving Retrieval-Augmented LMs with Compression and Selective Augmentation" by F. Xu, W. Shi, and E. Choi, ICLR 2024 [32]; "Learning to Filter Context for Retrieval-Augmented Generation" by Z. Wang, J. Araki, Z. Jiang, M. R. Parvez, and G. Neubig, 2023 [33].

Short note: Both reduce the retrieved context before it reaches the generator, by summarising or by filtering sentences.

Decision: Not used. These act after retrieval, while our work changes what is retrieved. Recorded here and listed in the references only.

Day 30 - 13 August 2025 - Long-context behaviour (brief)

Paper skimmed: "Lost in the Middle: How Language Models Use Long Contexts" by N. F. Liu et al., TACL 2024 [34].

Short note: Language models use information at the start and end of a long context more reliably than information placed in the middle, so simply retrieving more documents is not a solution.

Relevance in one line: Supports the argument that the top-k slots are a scarce resource and should not be spent on redundant documents, which is the practical justification for the novelty term. Not used further.

Day 31 - 14 August 2025 - Query-side methods (brief)

Papers skimmed: "Precise Zero-Shot Dense Retrieval without Relevance Labels" (HyDE) by L. Gao, X. Ma, J. Lin, and J. Callan, ACL 2023 [35]; "Query Rewriting for Retrieval-Augmented Large Language Models" by X. Ma, Y. Gong, P. He, H. Zhao, and N. Duan, EMNLP 2023 [36].

Short note: Both improve retrieval by changing the query - HyDE by generating a hypothetical answer document and searching with its embedding, and query rewriting by training a rewriter to produce better search queries.

Decision: Not used as components, but recorded as the main alternative approach to our problem. Important point noted for the report: these methods improve the query, while GAR changes the scoring function. If a reviewer asks why the gap cannot simply be turned into a better query, the answer is that MIGRES already does that, and its remaining failure mode is at the ranking stage. Listed in the references.

Day 32 - 15 August 2025 - Corpus organisation (brief)

Paper skimmed: "RAPTOR: Recursive Abstractive Processing for Tree-Organized Retrieval" by P. Sarthi et al., ICLR 2024 [37].

Short note: Documents are clustered and summarised recursively into a tree, so retrieval can happen at different levels of abstraction.

Decision: Not used. It changes the index, not the score, and it would break the fairness requirement that all compared systems use the same corpus. Listed in the references only.

Day 33 - 16 August 2025 - Consolidation: problem definition, research question and objectives

Work done: Closed the literature study and wrote the problem definition, the central research question, and the objectives, in the same wording used in the synopsis so that the documents stay consistent.

Problem definition recorded:

- In iterative RAG, conventional retrieval may repeatedly retrieve documents that are relevant to the original query but contain information that has already been obtained, which leads to redundant retrieval and reduces the usefulness of later retrieval steps.
- The information required to answer a multi-hop question changes as new evidence is collected, and existing retrieval methods may not sufficiently use the current missing information when selecting documents for the next step.
- Previously retrieved evidence can contain information that is already available to the system, so a retrieval mechanism is needed that considers the collected evidence together with the remaining information requirement.
- The effectiveness of using the current information gap and previously collected evidence for iterative retrieval needs to be experimentally investigated and compared with existing retrieval approaches.

Central research question recorded: Does explicitly modelling the current information gap, together with previously collected evidence, improve retrieval in iterative retrieval-augmented generation?

Objectives recorded:

- To investigate methods for reducing redundant retrieval of information that has already been obtained.
- To study and incorporate the current missing information as a factor in selecting documents during iterative retrieval.
- To investigate how previously collected evidence can be considered when selecting documents for subsequent retrieval steps.
- To evaluate the effectiveness of the developed retrieval approach against existing retrieval methods using suitable retrieval and question-answering measures.

Outcome: Literature phase closed. Mathematical formulation starts from the next day.

Day 34 - 17 August 2025 - Mathematical formulation, part 1: notation and representation

Work done: Fixed the notation and the representation function of GAR.

Recorded:

- At retrieval iteration t the inputs are the original question q, the evidence collected before the iteration E(t), the current missing-information requirements G(t) = {g1, ..., gm}, and the searchable corpus D. A candidate document is d, and an evidence item is e.

- A fixed embedding function phi maps any text to a 768-dimensional vector, using multi-qa-mpnet-base-cos-v1 [14][15]. The same function is used for the question, each gap requirement, each document and each evidence item, so that all comparisons happen in one space.
- Similarity is cosine similarity rescaled to the range 0 to 1: s(x, y) = 0.5 * (1 + cosine(phi(x), phi(y))). The rescaling is needed because the three components are later added with non-negative weights, so every component must lie on the same bounded scale.
- Document embeddings are computed once, before the experiment, and stored in the ANN index.

Outcome: Notation frozen and used consistently in all later sections.

Day 35 - 18 August 2025 - Mathematical formulation, part 2: gap relevance

Work done: Defined the first GAR component.

Recorded:

- Each gap requirement carries a non-negative weight w(i), with at least one weight positive. For the first implementation all weights are set to 1, which avoids introducing a second learned model.
- Weighted gap similarity: sG(d, G(t)) = sum over i of w(i) * s(d, g(i)), divided by the sum of the weights.
- Gap relevance combines the original question and the gap: R(d, q, G(t)) = lambda * s(d, q) + (1 - lambda) * sG(d, G(t)), with lambda between 0 and 1.
- Interpretation: lambda close to 1 emphasises the original question, lambda close to 0 emphasises the current gap. Keeping the question term is necessary because the gap text produced by the framework can be short or noisy, and dropping the question entirely would let retrieval drift away from the actual task.

Outcome: Component R defined.

Day 36 - 19 August 2025 - Mathematical formulation, part 3: gap coverage

Work done: Defined the second GAR component, following the coverage idea of [30].

Recorded:

- A fixed preprocessing pipeline produces a set of normalised content tokens for any text, written as tau(x). The pipeline (lowercasing, tokenisation, stop-word removal, normalisation) must be identical for documents and gap requirements, and must be fixed before the experiment.
- Coverage of a single requirement is normalised token recall: c(d, g(i)) = size of the intersection of tau(d) and tau(g(i)), divided by the size of tau(g(i)).
- Weighted coverage: C(d, G(t)) = sum over i of w(i) * c(d, g(i)), divided by the sum of the weights. The result lies between 0 and 1.

- If a requirement has an empty token set, it is invalid and is removed before retrieval, otherwise the division is undefined.

Reason for using tokens rather than embeddings here: Relevance already measures semantic closeness. Coverage is meant to measure whether the specific content of the requirement is actually present in the document, so a lexical measure gives a signal that is not a copy of the relevance term.

Outcome: Component C defined.

Day 37 - 20 August 2025 - Mathematical formulation, part 4: evidence novelty

Work done: Defined the third GAR component, following the redundancy argument of [28] and the prior art of MMR [18].

Recorded:

- Novelty against non-empty evidence: N(d, E(t)) = 1 - max over e in E(t) of s(d, e). A document that is very similar to some already collected evidence item receives a low novelty score; a document unlike all collected evidence receives a high score.
- The maximum is used rather than the mean, because a document is redundant if it repeats any single piece of evidence, not only if it repeats the evidence set on average.
- At the first iteration the evidence set is empty, and novelty is defined as 1, so that the first retrieval reduces to relevance and coverage only.

Risk recorded: Because novelty rewards dissimilarity, a completely unrelated document would receive a high novelty score. This is controlled by the weighted combination with relevance and coverage, and the weights are selected on the validation set. This trade-off must be examined explicitly in the ablation study.

Outcome: Component N defined.

Day 38 - 21 August 2025 - Mathematical formulation, part 5: final score and hyperparameters

Work done: Combined the three components and fixed the parameter set.

Recorded:

- Final score: S(d | q, E(t), G(t)) = alpha * R(d, q, G(t)) + beta * C(d, G(t)) + gamma * N(d, E(t)), with alpha, beta, gamma non-negative and alpha + beta + gamma = 1.
- Because each component lies between 0 and 1 and the weights form a convex combination, the final score also lies between 0 and 1. This makes scores comparable across iterations and across datasets.
- Scoring hyperparameters: alpha, beta, gamma, lambda. Retrieval hyperparameter: K0, the ANN candidate-pool size.
- The final retrieval depth k is an experimental setting and not a GAR parameter, and it must be identical for every compared system.

- The embedding model and the similarity function are configuration choices, not learned parameters. GAR therefore has no trainable parameters and no training stage.

Outcome: The GAR scoring function is complete.

Day 39 - 22 August 2025 - Candidate generation, ranking, and the architecture diagram

Work done: Separated the two stages of the algorithm and drew the architecture and flow diagrams for the methodology chapter.

Recorded:

- The candidate set is the union of the ANN results for the question and the ANN results for every gap requirement: C(t) = ANN(q, D) union the ANN results for each g(i), with duplicates removed. Searching with the gap requirements as well as the question is essential, because a document that answers only the current gap may not be near the question vector.
- The pipeline is: embedding, then ANN candidate generation, then GAR scoring, then top-k selection, then back to MIGRES for filtering, extraction and state update.
- Distinction recorded explicitly: if BM25 were used to build the candidate pool and GAR only reordered that pool, the experiment would be a GAR reranker experiment and not the full GAR retrieval experiment proposed here. Both can be reported, but they must not be confused.

Outcome: Figures 5.1 and 5.2 of the methodology drafted.

Day 40 - 23 August 2025 - Datasets, part 1

Papers studied: "Constructing Datasets for Multi-Hop Reading Comprehension Across Documents" (WikiHop) by J. Welbl, P. Stenetorp, and S. Riedel, TACL 2018 [38]; "HotpotQA: A Dataset for Diverse, Explainable Multi-Hop Question Answering" by Z. Yang et al., EMNLP 2018 [39]; "MuSiQue: Multihop Questions via Single-Hop Question Composition" by H. Trivedi, N. Balasubramanian, T. Khot, and A. Sabharwal, TACL 2022 [40].

Findings and suitability:

- WikiHop is built so that the answer requires combining information across documents, which tests whether retrieval can assemble complementary evidence instead of repeatedly retrieving one topic [38].
- HotpotQA provides sentence-level supporting facts and includes bridge and comparison questions. The FullWiki setting is used because it requires retrieval over a large Wikipedia corpus, and the official resources allow the retrieval component to be replaced by a custom IR system [39].
- MuSiQue is constructed by composing single-hop questions specifically to reduce shortcut solutions, with questions requiring two to four hops [40]. It is the hardest of the three for a query-only retriever and therefore the most informative for our hypothesis.

Relevance to our work: The sentence-level supporting-fact annotations in HotpotQA and the per-hop structure of MuSiQue are what make an evidence-coverage metric possible at all, since they tell us which evidence items should have been retrieved.

Outcome: Three datasets recorded with their retrieval settings.

Day 41 - 24 August 2025 - Datasets, part 2, and split policy

Papers studied: "Natural Questions: A Benchmark for Question Answering Research" by T. Kwiatkowski et al., TACL 2019 [41]; "TriviaQA: A Large Scale Distantly Supervised Challenge Dataset for Reading Comprehension" by M. Joshi, E. Choi, D. Weld, and L. Zettlemoyer, ACL 2017 [42]; "Did Aristotle Use a Laptop? A Question Answering Benchmark with Implicit Reasoning Strategies" (StrategyQA) by M. Geva et al., TACL 2021 [43].

Findings and suitability:

- Natural Questions contains real anonymised search queries with Wikipedia-based long and short answer annotations, and gives a single-hop control condition [41].
- TriviaQA pairs questions with independently collected evidence documents, so lexical overlap between question and evidence is lower and the evidence is heterogeneous [42].
- StrategyQA contains questions whose reasoning steps are implicit, with decompositions and Wikipedia evidence paragraphs, which tests gap-aware retrieval when the required steps are not stated in the question [43].

Split policy recorded: Where official splits exist they are used as published. The validation split is used only for GAR hyperparameter selection, and the test split only for the final evaluation. Gold answers and gold supporting evidence must never be used to build the gap state, the evidence state, or the candidate pool at test time.

Note recorded: Exact example counts must be taken from the downloaded release and written into the experiment configuration, since counts differ between releases and preprocessing scripts.

Outcome: All six datasets fixed, matching the datasets used in the MIGRES evaluation.

Day 42 - 25 August 2025 - Baseline setup and fairness controls

Work done: Fixed how each baseline will be run and what must be held constant across systems.

Recorded:

- Systems compared: MIGRES+BM25 [6] as the main baseline, MIGRES+DPR [7], MIGRES+Contriever [8], MIGRES+ColBERTv2 [10], MIGRES+SPLADE [11], and MIGRES+GAR as the proposed method.
- Every baseline is run from its official implementation and published checkpoint wherever possible. The exact checkpoint, index type and retrieval parameters of each baseline must be recorded, otherwise the comparison cannot be reproduced.
- Held constant across all systems: dataset split, question set, document corpus, final retrieval depth k, every MIGRES component outside retrieval, the answer-generation model, the stopping conditions, the evaluation scripts, and the random seeds where applicable.

- Only the retrieval algorithm changes. This is written as an explicit fairness statement in the methodology chapter.

Outcome: Baseline and fairness protocol fixed.

Day 43 - 26 August 2025 - Hyperparameter selection protocol

Paper studied: "Practical Bayesian Optimization of Machine Learning Algorithms" by J. Snoek, H. Larochelle, and R. P. Adams, NeurIPS 2012 [44].

Findings:

- Bayesian optimisation builds a probabilistic model of the objective from previous evaluations and uses it to choose the next configuration to try, which finds good settings in far fewer evaluations than grid search.
- This matters here because one evaluation is not a single retrieval call but a complete run of the MIGRES+GAR pipeline over the validation split, which is expensive.

Protocol recorded:

- The parameter set is alpha, beta, gamma, lambda and K0, searched under the constraints that alpha, beta and gamma are non-negative and sum to 1, that lambda lies between 0 and 1, and that K0 is drawn from a fixed discrete set, for example 50, 100, 200 and 500.
- Optimisation runs on the validation split through the complete pipeline, because GAR cannot be tuned in isolation: the evidence and gap states only exist when the framework is running.
- The objective is fixed before any test evaluation, and the primary objective is Recall@k on the validation split, which is the metric aligned with the main research question.
- The search space, the number of trials and the random seed are recorded before the run. After optimisation the configuration is frozen, and no test result is allowed to change it.

Outcome: Tuning protocol fixed and written into the methodology.

Day 44 - 27 August 2025 - Experimental design: research questions and hypotheses

Work done: Wrote the experiment plan.

Research questions recorded:

- RQ1: Does GAR improve retrieval quality over the MIGRES BM25 baseline?
- RQ2: Does GAR outperform strong dense, neural sparse and late-interaction retrieval methods?
- RQ3: Does replacing BM25 with GAR improve the downstream answer quality of MIGRES?
- RQ4: Which GAR component contributes most - relevance, coverage or novelty?
- RQ5: Does GAR generalise across all six datasets?
- RQ6: What is the computational cost of GAR compared with the baselines?
- RQ7: Does GAR improve retrieval without adding extra LLM calls?

Hypotheses recorded:

- Primary: null hypothesis, GAR does not improve retrieval performance over BM25; alternative, GAR improves retrieval performance over BM25.
- Secondary: null hypothesis, GAR is not better than the best baseline; alternative, GAR is better than the best baseline.
- Ablation: null hypothesis, removing a GAR component does not change performance; alternative, removing a component reduces performance.

Planned experiments recorded:

- Main comparison of the six systems on all six datasets.
- Component ablations: score without relevance, without coverage, and without novelty.
- Diagnostic variants: question-relevance only, and question plus gap relevance without novelty, to separate the benefit of adding the gap representation from the benefit of the full three-part score.
- Candidate-pool sensitivity over the K0 values, reported against both retrieval quality and latency.

Outcome: Experiment plan drafted. Additional experiments identified as missing are listed in Part B.

Day 45 - 28 August 2025 - Retrieval evaluation metrics

Papers studied: "Cumulated Gain-Based Evaluation of IR Techniques" by K. Jarvelin and J. Kekalainen, ACM TOIS, 2002 [45]; "Benchmarking Large Language Models in Retrieval-Augmented Generation" by J. Chen, H. Lin, X. Han, and L. Sun, AAAI 2024 [46].

Metrics recorded:

- Recall@k, the share of ground-truth evidence appearing in the top k results. This is the primary metric, because GAR is designed to retrieve the evidence that is still missing.
- Precision@k, the share of the retrieved set that is relevant.
- Mean Reciprocal Rank, which measures how early the first relevant result appears.
- Normalised Discounted Cumulative Gain at k, for graded relevance, using the discounting and normalisation defined in [45].
- Gap evidence coverage at k, an additional measure we report where annotations permit: the share of ground-truth evidence requirements retrieved within the top k. The mapping from each dataset's annotations to evidence requirements must be defined separately per dataset and written down before the experiment.

Note on [46]: The RGB benchmark tests noise robustness, negative rejection, information integration and counterfactual robustness in RAG. Read as a source of evaluation dimensions; noise robustness in particular is a property our novelty term could affect, and is a candidate for an additional experiment.

Outcome: Retrieval metric set fixed.

Day 46 - 29 August 2025 - Downstream metrics and statistical testing

Paper studied: "Statistical Significance Tests for Machine Translation Evaluation" by P. Koehn, EMNLP 2004 [47].

Recorded:

- Downstream answer quality is measured with the official metric of each benchmark: exact match, token-level F1, and accuracy for the classification-style dataset. Supporting-fact quality is reported where the annotations allow it. The official evaluation script is used whenever one is available.
- Because all systems are evaluated on the same questions, the comparison is paired. A paired bootstrap resampling procedure, as described in [47], or a paired permutation test, is used for per-question metrics.
- The exact test, the number of resamples and the significance level are fixed before the final test results are inspected.
- Effect sizes and confidence intervals are reported together with p-values, so that a small but significant difference is not presented as a large improvement.

Outcome: Statistical protocol fixed.

Day 47 - 30 August 2025 - Efficiency, cost, environment and reproducibility

Work done: Fixed what must be measured and what must be recorded.

Efficiency measures recorded: retrieval latency per question, ANN index build time, index memory, number of retrieval iterations per question, number of documents scored by GAR, number of LLM calls made by the whole MIGRES system, input and output token counts, and API cost where an external model is used.

Cost note: The cost per question is the input tokens times the input price plus the output tokens times the output price, averaged over the question set. The report must separate the local computation cost of GAR from the LLM cost of the surrounding framework, otherwise GAR would appear more expensive than it is.

Environment recorded: Python with Jupyter Notebook for experiments, PyTorch for model execution, Sentence Transformers for embeddings, a vector index for ANN search, a Bayesian optimisation library for tuning, and LaTeX for the report. Exact versions are written into the software table of the methodology chapter.

Reproducibility record fixed: dataset release and split, corpus version, retriever and checkpoint, GAR embedding model, final retrieval depth k, candidate-pool size K0, the selected alpha, beta, gamma and lambda, the optimisation seed, the ANN library and index type, hardware, software versions, and the exact MIGRES configuration including prompts and model settings.

Outcome: Reproducibility table drafted.

Day 48 - 31 August 2025 - Internal review and close of the methodology phase

Work done: Compiled the methodology chapter from Days 33 to 47 and reviewed it as a group against the synopsis.

Completed in this phase: literature study of 47 papers, selection of the base framework, the problem definition, the research question and objectives, the complete mathematical formulation of GAR, the architecture and flow diagrams, the dataset selection and split policy, the baseline and fairness protocol, the hyperparameter selection protocol, the experiment plan, the evaluation metrics, the statistical protocol, and the reproducibility record.

Review outcome: The chapter is complete as a plan, but three items are not yet settled and are recorded in Part B so that they are not forgotten: the MIGRES backbone model is not stated, the set of experiments is not yet complete, and a few reference details still need verification.

Decision on implementation: No code is written in this phase. Implementation starts on 1 January 2026, and the logbook continues from Part C on that date.

PART B - OPEN ITEMS CARRIED FORWARD

B.1 MIGRES backbone model not yet stated

The methodology claims that every MIGRES component outside retrieval is held fixed, but it does not yet state which language model MIGRES uses for missing-information identification, knowledge filtering, information extraction and answer solving. Until this is written down the fairness claim cannot be verified by a reader, and our own runs cannot be reproduced.

Action required: Take the backbone model, its version or checkpoint, the decoding settings (temperature, maximum tokens) and the prompt files from the experimental section of [5] and from the configuration in the official repository. Record them in the reproducibility table, and use the same settings for every compared system. If the reported backbone is not available to us, record the substitute model explicitly and state that the absolute numbers are therefore not directly comparable with the published MIGRES results, while the relative comparison between retrievers remains valid because all systems share the same substitute.

B.2 Experiment set is incomplete

The current plan covers the main comparison, the component ablations, and candidate-pool sensitivity. The following experiments are needed before the evaluation can answer the research questions fully, and are to be added to the methodology chapter.

- Retrieval depth sensitivity: vary k and report whether the advantage of GAR grows or shrinks as more documents are returned, since a large k can hide a ranking improvement.
- Iteration-wise analysis: report the metrics per retrieval iteration, not only as a single average. GAR should show little or no benefit at the first iteration, where the evidence set is empty, and a growing benefit at later iterations. This is the most direct test of the central claim and it is currently missing.
- Redundancy measurement: report the overlap between newly retrieved documents and already collected evidence for each system, since reducing redundancy is the first objective and no metric currently measures it directly.
- Gap-quality sensitivity: run GAR with the gap produced by the framework, with a perturbed or truncated gap, and with an oracle gap derived from the annotations for analysis only. This separates the contribution of GAR from the quality of the gap that MIGRES supplies.

- Embedding-model sensitivity: repeat with at least one alternative sentence-embedding model, to show that the result is a property of the scoring function and not of one particular encoder.
- GAR as a reranker versus GAR as a full retriever: run GAR over a BM25 candidate pool as a separate condition, so that the effect of the candidate generation stage is separated from the effect of the scoring function.
- Number of hops: break the results down by the number of reasoning hops required, since the benefit is expected to appear on questions with more hops.
- Failure analysis: sample questions where GAR performs worse than the baseline and categorise the causes, in particular cases where a high novelty score promotes an unrelated document.
- Backbone sensitivity: repeat the main comparison with at least one additional generation model, if compute allows, to show the result is not tied to one LLM.
- Statistical reporting: confirm that every reported comparison carries an effect size and a confidence interval, not only a significance marker.

B.3 Reference details to verify before submission

The publication venue and page numbers of a small number of entries in Part D must be checked against the official proceedings before the final report is submitted, since several of these papers exist both as preprints and as published versions. The entries flagged during the study are the Contriever paper [8], the SPLADE family [11], where the first version and the second version are different papers, and the demonstrate-search-predict entry [27].

PART C - IMPLEMENTATION PHASE (NOT YET STARTED)

Implementation has not been started in this phase. The work planned below begins on 1 January 2026 and will be logged in the same day-wise format in this logbook. The order below is the order recorded on Day 47 as the complete experimental flow.

- Prepare the official dataset splits and build the searchable corpus.
- Reproduce MIGRES with its BM25 retriever and match the published baseline numbers as closely as the environment permits.
- Verify the MIGRES retrieval interface and confirm exactly how the evidence state and the gap state are exposed at the point of the retrieval call.
- Build the fixed GAR document embeddings for the corpus.
- Build the ANN index.
- Implement the GAR equations and unit-test each component separately against hand-computed examples.
- Run Bayesian optimisation on the validation split through the complete pipeline.
- Freeze the selected alpha, beta, gamma, lambda and K0.
- Run the held-out test split with the frozen configuration.

- Run the five baseline retrievers using their official implementations.
- Evaluate retrieval quality, downstream answer quality, efficiency and cost.
- Run the ablations and the additional experiments listed in B.2 under the same frozen protocol.
- Analyse the results, run the statistical tests, and write the results chapter.

Space is left below for the daily entries of the implementation phase.

Day ___ - ___ / ___ / 2026 - _____

Day ___ - ___ / ___ / 2026 - _____

Day ___ - ___ / ___ / 2026 - _____

Day ___ - ___ / ___ / 2026 - _____

Day ___ - ___ / ___ / 2026 - _____

Day ___ - ___ / ___ / 2026 - _____

Day ___ - ___ / ___ / 2026 - _____

Day ___ - ___ / ___ / 2026 - _____

PART D - REFERENCES

[1] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Kuttler, M. Lewis, W. Yih, T. Rocktaschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in Advances in Neural Information Processing Systems, vol. 33, 2020, pp. 9459-9474.

[2] K. Guu, K. Lee, Z. Tung, P. Pasupat, and M.-W. Chang, "REALM: Retrieval-augmented language model pre-training," in Proceedings of the 37th International Conference on Machine Learning, 2020, pp. 3929-3938.

[3] G. Izacard and E. Grave, "Leveraging passage retrieval with generative models for open domain question answering," in Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics, 2021, pp. 874-880.

[4] Y. Gao, Y. Xiong, X. Gao, K. Jia, J. Pan, Y. Bi, Y. Dai, J. Sun, M. Wang, and H. Wang, "Retrieval-augmented generation for large language models: A survey," arXiv preprint arXiv:2312.10997, 2023.

[5] K. Wang, F. Duan, P. Li, S. Wang, and X. Cai, "LLMs know what they need: Leveraging a missing information guided framework to empower retrieval-augmented generation," in Proceedings of the 31st International Conference on Computational Linguistics, 2025, pp. 2379-2400.

[6] S. Robertson and H. Zaragoza, "The probabilistic relevance framework: BM25 and beyond," Foundations and Trends in Information Retrieval, vol. 3, no. 4, pp. 333-389, 2009.

[7] V. Karpukhin, B. Oguz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W. Yih, "Dense passage retrieval for open-domain question answering," in Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing, 2020, pp. 6769-6781.

[8] G. Izacard, M. Caron, L. Hosseini, S. Riedel, P. Bojanowski, A. Joulin, and E. Grave, "Unsupervised dense information retrieval with contrastive learning," Transactions on Machine Learning Research, 2022.

[9] O. Khattab and M. Zaharia, "ColBERT: Efficient and effective passage search via contextualized late interaction over BERT," in Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval, 2020, pp. 39-48.

[10] K. Santhanam, O. Khattab, J. Saad-Falcon, C. Potts, and M. Zaharia, "ColBERTv2: Effective and efficient retrieval via lightweight late interaction," in Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics, 2022, pp. 3715-3734.

[11] T. Formal, B. Piwowarski, and S. Clinchant, "SPLADE: Sparse lexical and expansion model for first stage ranking," in Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval, 2021, pp. 2288-2292.

[12] R. Nogueira and K. Cho, "Passage re-ranking with BERT," arXiv preprint arXiv:1901.04085, 2019.

[13] R. Nogueira, Z. Jiang, R. Pradeep, and J. Lin, "Document ranking with a pretrained sequence-to-sequence model," in Findings of the Association for Computational Linguistics: EMNLP 2020, 2020, pp. 708-718.

[14] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using siamese BERT-networks," in Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing, 2019, pp. 3982-3992.

[15] Sentence Transformers, "multi-qa-mpnet-base-cos-v1," Hugging Face model card, 2020, 768-dimensional sentence-transformers model designed for semantic search.

[16] J. Johnson, M. Douze, and H. Jegou, "Billion-scale similarity search with GPUs," arXiv preprint arXiv:1702.08734, 2017.

[17] Y. A. Malkov and D. A. Yashunin, "Efficient and robust approximate nearest neighbor search using hierarchical navigable small world graphs," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 42, no. 4, pp. 824-836, 2020.

[18] J. Carbonell and J. Goldstein, "The use of MMR, diversity-based reranking for reordering documents and producing summaries," in Proceedings of the 21st Annual International ACM SIGIR Conference on Research and Development in Information Retrieval, 1998, pp. 335-336.

[19] N. Thakur, N. Reimers, A. Ruckle, A. Srivastava, and I. Gurevych, "BEIR: A heterogeneous benchmark for zero-shot evaluation of information retrieval models," in Advances in Neural Information Processing Systems, Datasets and Benchmarks Track, 2021.

[20] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, "Self-RAG: Learning to retrieve, generate, and critique through self-reflection," in International Conference on Learning Representations, 2024.

[21] S. Jeong, J. Baek, S. Cho, S. J. Hwang, and J. C. Park, "Adaptive-RAG: Learning to adapt retrieval-augmented large language models through question complexity," in Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics, 2024, pp. 7036-7050.

[22] H. Trivedi, N. Balasubramanian, T. Khot, and A. Sabharwal, "Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions," in Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics, 2023, pp. 10014-10037.

[23] O. Press, M. Zhang, S. Min, L. Schmidt, N. A. Smith, and M. Lewis, "Measuring and narrowing the compositionality gap in language models," in Findings of the Association for Computational Linguistics: EMNLP 2023, 2023, pp. 5687-5711.

[24] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao, "ReAct: Synergizing reasoning and acting in language models," in International Conference on Learning Representations, 2023.

[25] Z. Jiang, F. F. Xu, L. Gao, Z. Sun, Q. Liu, J. Dwivedi-Yu, Y. Yang, J. Callan, and G. Neubig, "Active retrieval augmented generation," in Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, 2023, pp. 7969-7992.

[26] Z. Shao, Y. Gong, Y. Shen, M. Huang, N. Duan, and W. Chen, "Enhancing retrieval-augmented large language models with iterative retrieval-generation synergy," in Findings of the Association for Computational Linguistics: EMNLP 2023, 2023, pp. 9248-9274.

[27] O. Khattab, K. Santhanam, X. L. Li, D. Hall, P. Liang, C. Potts, and M. Zaharia, "Demonstrate-search-predict: Composing retrieval and language models for knowledge-intensive NLP," arXiv preprint arXiv:2212.14024, 2022.

[28] M. Pickett, J. Hartman, A. K. Bhowmick, R. ul Alam, and A. Vempaty, "Better RAG using relevant information gain," arXiv preprint arXiv:2407.12101, 2024.

[29] Z. Wang, Z. Liang, Z. Shao, Y. Ma, H. Dai, B. Chen, L. Mao, C. Lei, Y. Ding, and H. Li, "InfoGain-RAG: Boosting retrieval-augmented generation through document information gain-based reranking and filtering," in Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, 2025, pp. 7190-7204.

[30] V. Yadav, S. Bethard, and M. Surdeanu, "Unsupervised alignment-based iterative evidence retrieval for multi-hop question answering," in Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, 2020, pp. 4514-4525.

[31] J. Hwang, J. Park, H. Park, D. Kim, S. Park, and J. Ok, "Retrieval-augmented generation with estimation of source reliability," in Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, 2025, pp. 34279-34303.

[32] F. Xu, W. Shi, and E. Choi, "RECOMP: Improving retrieval-augmented LMs with compression and selective augmentation," in International Conference on Learning

Representations, 2024.

[33] Z. Wang, J. Araki, Z. Jiang, M. R. Parvez, and G. Neubig, "Learning to filter context for retrieval-augmented generation," arXiv preprint arXiv:2311.08377, 2023.

[34] N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang, "Lost in the middle: How language models use long contexts," Transactions of the Association for Computational Linguistics, vol. 12, pp. 157-173, 2024.

[35] L. Gao, X. Ma, J. Lin, and J. Callan, "Precise zero-shot dense retrieval without relevance labels," in Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics, 2023, pp. 1762-1777.

[36] X. Ma, Y. Gong, P. He, H. Zhao, and N. Duan, "Query rewriting for retrieval-augmented large language models," in Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, 2023, pp. 5303-5315.

[37] P. Sarthi, S. Abdullah, A. Tuli, S. Khanna, A. Goldie, and C. D. Manning, "RAPTOR: Recursive abstractive processing for tree-organized retrieval," in International Conference on Learning Representations, 2024.

[38] J. Welbl, P. Stenetorp, and S. Riedel, "Constructing datasets for multi-hop reading comprehension across documents," Transactions of the Association for Computational Linguistics, vol. 6, pp. 287-302, 2018.

[39] Z. Yang, P. Qi, S. Zhang, Y. Bengio, W. Cohen, R. Salakhutdinov, and C. D. Manning, "HotpotQA: A dataset for diverse, explainable multi-hop question answering," in Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing, 2018, pp. 2369-2380.

[40] H. Trivedi, N. Balasubramanian, T. Khot, and A. Sabharwal, "MuSiQue: Multihop questions via single-hop question composition," Transactions of the Association for Computational Linguistics, vol. 10, pp. 539-554, 2022.

[41] T. Kwiatkowski, J. Palomaki, O. Redfield, M. Collins, A. Parikh, C. Alberti, D. Epstein, I. Polosukhin, J. Devlin, K. Lee, K. Toutanova, L. Jones, M. Kelcey, M.-W. Chang, A. M. Dai, J. Uszkoreit, Q. Le, and S. Petrov, "Natural questions: A benchmark for question answering research," Transactions of the Association for Computational Linguistics, vol. 7, pp. 452-466, 2019.

[42] M. Joshi, E. Choi, D. Weld, and L. Zettlemoyer, "TriviaQA: A large scale distantly supervised challenge dataset for reading comprehension," in Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics, 2017, pp. 1601-1611.

[43] M. Geva, D. Khashabi, E. Segal, T. Khot, D. Roth, and J. Berant, "Did Aristotle use a laptop? A question answering benchmark with implicit reasoning strategies," Transactions of the Association for Computational Linguistics, vol. 9, pp. 346-361, 2021.

[44] J. Snoek, H. Larochelle, and R. P. Adams, "Practical Bayesian optimization of machine learning algorithms," in Advances in Neural Information Processing Systems, vol. 25, 2012.

[45] K. Jarvelin and J. Kekalainen, "Cumulated gain-based evaluation of IR techniques," ACM Transactions on Information Systems, vol. 20, no. 4, pp. 422-446, 2002.

[46] J. Chen, H. Lin, X. Han, and L. Sun, "Benchmarking large language models in retrieval-augmented generation," in Proceedings of the AAAI Conference on Artificial Intelligence, vol. 38, 2024, pp. 17754-17762.

[47] P. Koehn, "Statistical significance tests for machine translation evaluation," in Proceedings of the 2004 Conference on Empirical Methods in Natural Language Processing, 2004, pp. 388-395.